

Suraj Sah Kanu

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Professional Summary

Machine Learning Engineer with research experience in **Deep Learning**, **Computer Vision**, and **Medical Image Analysis**. Proven track record implementing state-of-the-art architectures (U-Net, ResNet, YOLOv8) for clinical imaging tasks, achieving **92%+ validation accuracy**. Skilled in neural network optimization, transfer learning, MLOps pipelines, and deploying scalable AI solutions with Docker and AWS. Experienced in end-to-end model development from data annotation and preprocessing through production deployment.

Core Competencies

Machine Learning • Deep Learning • Computer Vision • Medical Image Analysis • Transfer Learning • Model Optimization • Feature Engineering • Supervised & Unsupervised Learning • Hyperparameter Tuning • Retrieval-Augmented Generation (RAG) • Generative AI • MLOps • CI/CD • Data Augmentation

Technical Skills

Programming Languages: Python, SQL

ML & DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, MONAI, Ultralytics YOLOv8, OpenCV

MLOps & Deployment: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, AWS SageMaker, Amazon Bedrock, FastAPI, Streamlit

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI

Medical Imaging: DICOM, NIfTI, 3D Image Processing, Image Segmentation, LabelStudio, ITK-SNAP, 3D Slicer

Databases: MySQL, PostgreSQL, SQLite

Developer Tools: Git, GitHub, Jupyter, Google Colab, VS Code, PyCharm

Experience

Research Intern – Machine Learning

Aug 2024 – Apr 2026

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII) – MAPMED Lab

Kathmandu, Nepal

- Built end-to-end **medical image embedding extraction pipeline** processing 3D volumetric CT/MRI scans using MedicalNet ResNet models; implemented global and ROI-based embeddings across 4 feature layers (256–2048 dims).
- Engineered **3 fine-tuning strategies** (partial, full, linear probe) for automated esophagitis grading using the RTOG dataset, achieving **92%+ validation accuracy** via stratified k-fold cross-validation.
- Developed advanced **3D preprocessing pipeline** with center-crop normalization, multi-scale augmentation, and NIfTI format conversion, reducing image processing overhead by **65%** through batch automation.
- Created comprehensive **embedding analysis framework** with PCA dimensionality reduction, contrastive similarity metrics, Isolation Forest outlier detection, and 16+ automated visualizations for quality assurance.
- Designed robust data utilities including CSV label merging, DICOM/NIfTI validation, cross-platform logging, and production-ready error handling; documented **15+ technical workflows** for reproducible research.

Projects

YOLOv8 Brain Tumor Detection System | *Deep Learning, FastAPI, Transfer Learning, LabelStudio*

[github](#)

- Developed production-grade **FastAPI** backend serving YOLOv8 for real-time brain MRI tumor detection, processing **50+ images/minute** with **100ms latency**.
- Curated and hand-annotated **5,000+ brain MRI images** in LabelStudio across 3 tumor classes (Glioma, Meningioma, No Tumor) with 95%+ annotation quality.
- Achieved **92.3% mAP@0.5** via custom hyperparameter tuning, multi-scale augmentation, and k-fold cross-validation on Google Colab T4 GPU.
- Implemented transfer learning from COCO weights with custom anchor boxes and multi-scale training, improving small tumor detection by **18%**; deployed scalable REST API with batch processing (32 images/batch) and model versioning.

NeuroScan AI – Brain Tumor Classification | *TensorFlow, MobileNetV2, Docker, Grad-CAM*

[github](#)

- Built end-to-end classification system achieving **94.3% accuracy** on 3,000+ MRI scans using MobileNetV2 with transfer learning and fine-tuning.
- Designed interactive Plotly dashboard with 2D/3D MRI visualizations and **Grad-CAM** explainability heatmaps, improving clinical trust by 40%.
- Containerized with **Docker**, reducing deployment complexity by 90% and enabling cross-platform deployment with **50ms inference time**.
- Implemented preprocessing pipeline (skull stripping, normalization, augmentation) improving model robustness by 12% across diverse scanner types.

HealthGuide Assistant AI | *Amazon Bedrock, LLM, RAG, Generative AI, PartyRock*

[Live Demo](#)

- Developed intelligent **healthcare chatbot** using Amazon Bedrock (Claude & Llama foundation models), serving 500+ user queries with 85% satisfaction rate.

- Implemented **RAG (Retrieval-Augmented Generation)** with custom medical knowledge base, improving response accuracy by 30% and reducing hallucinations by 45%.
- Selected as finalist in **AWS Generative AI Builder Challenge** during AWS AI/ML Scholarship program, demonstrating practical LLM applications in healthcare.

Education

Bachelor in Computer Application

National Infotech College, Tribhuvan University

Nepal

Higher Secondary Education (Science)

Birgunj Public College

Birgunj, Nepal

Achievements & Certifications

AWS AI/ML Scholar 2025 – Selected among **top 3%** of applicants through competitive AWS AI & ML Scholarship Program based on technical excellence.

AWS Certified: Machine Learning Foundations – Hands-on expertise in ML principles, AWS services, and cloud-based model training.

AWS Certified: Generative AI with AWS – Practical experience with Amazon Bedrock, foundation models, and generative AI application development.

Positions of Responsibility

Teaching Assistant

2025

Annual Nepal AI School

Nepal

- Supported instructors in delivering **AI and data science** sessions; assisted students with hands-on labs and doubt resolution, evaluated assignments, and coordinated workshops and technical events.